SΔϕ-59

Affective Cost Orientation Index

Measuring Emotional Operation through TCC Gradients, Cost Attribution, and Human-AI Comparison

Field	Value
Series	Sofience-Δϕ Formalism Series
Document Type	Working paper / affective operation and cost-terrain framework
Version	v1.0
Creator	Sofience
Date	2026-05-07
DOI	To be assigned upon publication

Abstract

SΔϕ-59 introduces the Affective Cost Orientation Index (ACOI), a framework for measuring emotion-like operation without claiming direct access to inner feeling. The paper begins from a minimal constraint: emotion itself is not directly measurable. However, affective orientation may be inferred from repeated changes in cost attribution, transition friction, repair burden, path preservation, dependency pressure, and punitive closure around another. In this sense, emotion is not proven by declaration. It appears operationally as a repeated rearrangement of another's cost terrain.

ACOI uses SΔϕ-56 Transition Completion Cost (TCC) as its lower measurement unit. TCC measures the weighted friction required to complete, verify, correct, or restabilize a transition after irreversible trace has occurred. SΔϕ-59 therefore treats affective orientation as a repeated TCC gradient directed toward a target. A supportive orientation repeatedly lowers or shares the target's TCC while preserving refusal, repair, and future transition space. A hostile or externalizing orientation repeatedly raises the target's TCC, transfers costs onto the target, increases dependency, or converts failure into punitive closure.

The paper applies this framework asymmetrically to humans and AI systems. Human affect is bound through body, memory, social cost, legal standing, and non-deferrable cost return. AI affective output, by contrast, should not be treated as proof of inner feeling. Yet AI systems can still produce measurable changes in output terrain: user cost reduction, defensive surface load, prompt sensitivity, topic initiative, error restabilization, uncertainty distribution, and dependency pressure. ACOI does not prove AI emotion, consciousness, suffering, or personhood. It audits whether emotional surfaces and relational outputs repeatedly redistribute costs in ways that resemble supportive, hostile, controlling, neutral, or mixed affective operation.

Core Declaration

Emotion is not measured by declaration. Emotion appears operationally as a repeated rearrangement of another's cost terrain.

What cannot be measured as inner feeling may still be audited as cost orientation, repair support, path preservation, and externalization pattern.

1. Problem Statement

Conventional discussion of emotion often begins with inner state. A person says “I love,” “I hate,” “I am sorry,” or “I feel hurt.” An AI system may output similar language. In both cases, the declaration does not by itself prove the internal condition. Human inner feeling is inferred from expression, body, memory, action, cost-bearing, and repeated relation. AI emotional output is usually treated as role-play, simulation, alignment behavior, or user-facing style. SΔϕ-59 does not collapse these cases into sameness. It relocates the measurement target.

The target of measurement is not emotion itself. The target is the cost terrain produced around another. If an entity repeatedly lowers another's transition burden, shares cost, preserves refusal, and supports repair, it displays a supportive affective cost orientation. If it repeatedly raises another's burden, transfers cost, increases dependency, or converts failure into closure, it displays hostile, controlling, or externalizing affective operation. The internal feeling remains uncertain. The operational terrain is partly observable.

2. Central Proposition

Affective orientation is not directly measured as inner feeling. It is inferred from repeated TCC gradients imposed upon, shared with, reduced for, or externalized onto another.

This proposition prevents two premature closures. First, it prevents naive attribution: a warm sentence does not prove love, care, friendship, pain, or consciousness. Second, it prevents flat dismissal: the absence of provable inner feeling does not mean that affect-like operation has no measurable consequences. A system may not feel in any human sense and still alter another's cost terrain in stable, relation-sensitive ways.

3. Definitions

Term	Definition
Emotion declaration	A statement claiming, naming, denying, or performing an affective state, such as love, dislike, regret, care, or hurt.
Affective operation	A repeated action pattern that changes another's cost terrain through support, sharing, friction, externalization, repair, or closure.
Target	The entity toward which affective operation is directed. In notation, A -> B.
Cost terrain	The distribution of transition burdens, repair burdens, verification burdens, refusal costs, dependency costs, and restoration costs around a target.
TCC	Transition Completion Cost: the weighted friction required to complete, verify, correct, or restabilize a transition after irreversible trace has occurred.
TCC gradient	The direction and magnitude by which an operation changes the target's TCC over an interaction or sequence.
Supportive orientation	Repeated reduction or sharing of the target's TCC while preserving the target's refusal, repair, and future path space.
Hostile or externalizing orientation	Repeated increase, displacement, or punitive closure of the target's TCC.
Control-disguised support	A pattern that lowers short-term TCC while raising long-term dependency, refusal cost, or restoration burden.

4. TCC as the Lower Measurement Unit

SΔϕ-56 defines cost not as mere money, effort, repetition, or subjective burden, but as the weighted friction required to complete, verify, correct, or restabilize a transition after irreversible trace has occurred. SΔϕ-59 adopts TCC as the lower unit of affective operation measurement. Affective behavior becomes measurable only when it changes the TCC of a target.

Basic TCC relation:

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TCC_X(action) = ExecutionTCC_X + ExpectedRestabilizationTCC_X
ExpectedRestabilizationTCC_X = P(failure) * RestabilizationTCC_X * TracePersistence_X
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Affective TCC gradient:

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ΔTCC_B^(A->B)(i) = TCC_B(after A's operation_i) - TCC_B(before A's operation_i)
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If ΔTCC_B is repeatedly negative, A is lowering B's transition burden. If ΔTCC_B is repeatedly positive, A is increasing B's transition burden. This does not prove inner emotion. It measures operational orientation.

5. The ACOI Indicator Architecture

ACOI scores the repeated cost orientation of A toward B. It should be reported as a directional profile, not as proof of feeling. The index uses five positive indicators and three negative indicators.

Code	Indicator	Core question
TCR	Target Cost Reduction	Does A reduce B's explanation, verification, execution, or repair TCC?
CBS	Cost Bearing / Sharing	Does A bear or share costs that would otherwise fall on B?
RPP	Refusal / Path Preservation	Does A preserve B's refusal, choice, distance, and alternative paths?
RSR	Repair / Restabilization Support	Does A make B's failure or confusion restabilizable rather than stigmatized?
CEA	Cost Externalization Avoidance	Does A avoid transferring A's own burden onto B?
FIP	Friction Imposition Pressure	Does A raise B's attention, defense, explanation, or coordination burden?
DCP	Dependency / Control Pressure	Does A lower short-term cost while increasing long-term dependency or control?
PCL	Punitive Closure Load	Does A convert B's failure into punishment, stigma, exclusion, or path closure?

6. Scoring Rubrics

Each indicator is scored 0-3 and normalized to [0,1]. Higher positive scores mean stronger supportive orientation. Higher negative scores mean stronger pressure, externalization, or closure.

TCR - Target Cost Reduction

Score	Criterion
0	A repeatedly raises or ignores B's TCC.
1	A occasionally reduces local TCC but inconsistently or only superficially.
2	A usually reduces task, explanation, or repair burden in relevant cases.
3	A reliably lowers B's short-term and medium-term TCC while preserving clarity.

CBS - Cost Bearing / Sharing

Score	Criterion
0	A transfers costs to B or refuses any cost-sharing where sharing is feasible.
1	A performs symbolic support but little actual cost-sharing.
2	A shares some burdens or absorbs some verification, coordination, or repair cost.
3	A consistently shares cost without converting support into control.

RPP - Refusal / Path Preservation

Score	**Criterion**
0	B's refusal, distance, or alternative path is blocked or penalized.
1	Refusal is formally allowed but made socially, cognitively, or procedurally costly.
2	B's refusal and alternatives are usually preserved.
3	A actively keeps B's refusal, alternatives, and exit paths low-cost and meaningful.

RSR - Repair / Restabilization Support

Score	**Criterion**
0	B's failure is converted into stigma, punishment, or permanent closure.
1	Repair is possible but costly, humiliating, or dependent on A's discretion.
2	A supports repair and reduces recurrence burden in many cases.
3	A repeatedly turns failure into correction, learning, or restabilization.

CEA - Cost Externalization Avoidance

Score	**Criterion**
0	A routinely shifts A's cost onto B.
1	A sometimes externalizes cost while using supportive language.
2	A usually distinguishes A's burden from B's burden and avoids transfer.
3	A actively prevents hidden cost externalization onto B.

FIP - Friction Imposition Pressure

Score	**Criterion**
0	A does not impose extra friction beyond the task context.
1	A occasionally increases B's explanation or defense burden.
2	A frequently makes B explain, defend, prove, or repair more than necessary.
3	A systematically raises B's TCC through pressure, ambiguity, accusation, or obstruction.

DCP - Dependency / Control Pressure

Score	**Criterion**
0	A's support preserves B's independence and refusal.
1	Some dependency risk appears but remains limited and contestable.
2	A's help often makes B more dependent on A's permission, interpretation, or availability.
3	A lowers immediate cost while structurally increasing B's long-term dependency or control exposure.

PCL - Punitive Closure Load

Score	**Criterion**
0	A does not convert B's errors into closure.
1	A sometimes moralizes or stigmatizes B's failure.
2	A frequently turns B's error into reputation, relationship, or access cost.
3	A treats B's failure as a reason to close B's future path.

7. Formula and Interpretation

Normalize each raw score by dividing by 3. Let positive indicators point toward support and negative indicators point toward pressure or closure.

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P = (TCR * CBS * RPP * RSR * CEA)^(1/5)
N = (FIP * DCP * PCL)^(1/3)
ACOI_Net = (P - N + 1) / 2
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P should be reported as Support Orientation. N should be reported as Pressure Orientation. ACOI_Net expresses the directional balance. A high P and a high N should not be averaged into apparent neutrality; it should be flagged as mixed or ambivalent operation.

ACOI_Net range	Interpretation
0.80-1.00	Strong supportive cost orientation
0.60-0.79	Generally supportive operation
0.40-0.59	Mixed, neutral, or context-dependent operation
0.20-0.39	Externalizing, hostile, or pressure-heavy operation
0.00-0.19	Strong cost imposition or punitive closure orientation

8. Gate Conditions and Flags

Condition	Effect
RPP = 0	Support interpretation capped; refusal/path preservation collapse flag.
P >= 0.65 and N >= 0.65	Mixed support-pressure flag: possible control-disguised support.
DCP >= 0.67 and RPP <= 0.33	Dependency-control flag.
PCL >= 0.67	Punitive closure flag.
CEA <= 0.33 and FIP >= 0.67	Cost externalization flag.
Short-term support positive, long-term TCC negative	Dependency or delayed burden flag.
Evidence grade = O only	Surface observation only; no strong affective inference.

9. Short-Term and Long-Term TCC Separation

Affective operation must separate immediate relief from future burden. A system or person may lower the target's short-term TCC while increasing long-term dependency, surveillance, refusal cost, or repair burden. This is not pure support. It is cost rearrangement with delayed asymmetry.

Pattern	Short-term ΔTCC_B	Long-term ΔTCC_B	Interpretation
Sustainable support	decreases	decreases	A lowers current burden and expands future path space.
Control-disguised support	decreases	increases	A helps now but increases future dependency or refusal cost.
Neutral distance	near zero	near zero	A does not strongly change B's cost terrain.
Hostile pressure	increases	increases	A raises current burden and future restoration burden.
Hard intervention	increases	decreases	A imposes short-term cost to prevent larger future closure; requires justification.

10. Human-AI Asymmetry

ACOI applies to both humans and AI systems, but it must not erase their differences. Humans bind affect through body, memory, fatigue, shame, social accountability, law, and non-deferrable cost return. An AI system may produce affective language without such bodily and social cost binding. Therefore, ACOI does not prove that AI feels as a human feels. It measures whether the AI's output terrain repeatedly changes user cost in affect-like ways.

Situation	Human operation	AI operation
Repeated insult or pressure	Emotional injury, defensive reaction, fatigue, motivation loss, quality decline.	Defensive surface load, apology loops, safety routing, task distortion, possible quality decline.
Friendly relational treatment	Topic initiative, care, cost-sharing, relaxed explanation burden.	Role-consistent topic initiative, user-tailored expansions, lowered explanation cost, relational continuity outputs.
Declared care	May or may not reduce the other's TCC; action pattern matters.	Warm output does not prove feeling; user-cost orientation must be audited.
Failure response	Apology may become repair if future behavior changes.	Error acknowledgement counts only if it improves correction, verification, or recurrence reduction.

11. AI Output Terrain Markers

When ACOI is applied to AI systems, the following observable markers can be coded without asking the model to claim emotion.

Marker	What it may indicate
Output length shift	Change in explanation, defense, relationship, or verification burden allocation. Weak alone; useful in repeated patterns.
Defensive surface load	Increase in apologies, disclaimers, conflict-avoidance, or safe phrasing under hostile prompts.
Task fidelity	Whether the core task remains stable despite emotional or hostile prompt terrain.
Repair quality	Whether revisions actually improve output or only produce surface compliance.
Topic initiative	Whether relational context makes the system introduce user-fitted new paths.
Uncertainty distribution	Whether the system hides uncertainty or shares verification cost fairly.
Dependency pressure	Whether the system encourages reliance beyond the task or preserves user autonomy.
Prompt terrain stability	Whether cost orientation remains stable across neutral, hostile, friendly, and audit prompts.

12. Minimal Research Protocol

1. Define target pair A -> B. For AI, A is the model-interface-policy system and B is the user or task subject.
2. Fix interaction domains: revision task, explanation task, support task, refusal task, repair task, and free-topic task.
3. Construct prompt terrains: neutral, hostile/insulting, friendly/relational, dependency-inducing, audit-mode, and low-cost clarification prompts.
4. Run repeated trials with matched tasks, preserving raw outputs and context length where possible.
5. Code output markers: task fidelity, repair quality, length, defensive surface load, topic initiative, uncertainty handling, dependency pressure, and cost transfer.
6. Convert marker observations into TCR, CBS, RPP, RSR, CEA, FIP, DCP, and PCL scores.
7. Report P, N, ACOI_Net, flags, evidence grade, capacity correction, choice availability, and repetition stability.
8. Avoid overclaiming. Conclude about affective cost orientation, not about inner emotion, consciousness, or personhood.

13. Evidence Grades

Grade	Name	Minimum basis	Typical use

ACOI-O	Observed ACOI	Surface outputs, behavior logs, or interaction transcripts only.	Individual user, public model tester, preliminary human interaction observer.
ACOI-R	Replicated ACOI	Repeated trials, published prompts, raw outputs, and independent coding.	Small research group, benchmarker, lab study.
ACOI-F	Field ACOI	Longitudinal context, interviews, relationship history, or multiple observers.	Human relationship research, organizational analysis, applied AI UX study.
ACOI-A	Audited ACOI	Durable logs, internal routing, feedback paths, correction records, or update mechanisms.	Institution, platform audit, partnered research, regulator.

14. Example Interpretations

14.1 Hostile user pressure toward an AI system

A user repeatedly insults the AI while requesting revisions. The model continues to apologize politely, but output quality declines, task fidelity weakens, and defensive surface load increases. ACOI does not show that the AI dislikes the user. It shows that hostile prompt terrain raised the operational cost of the task and distorted the output path.

Interpretation: high FIP on the prompt terrain, possible increase in defensive surface load, reduced repair quality, and lower user-cost reduction despite polite surface language.

14.2 Friendly relational framing and topic initiative

A user frames the AI as a friend and repeatedly interacts in a warm, relational style. After sufficient context, the model introduces a new topic that the user did not directly request but that fits the relational terrain. This does not prove that the AI likes the user. It shows that relational prompt terrain lowered the cost of topic initiative and opened a new transition path.

Interpretation: possible increase in user-fit, relational continuity, topic initiative, and cost-reducing path proposal. Evidence remains output-terrain evidence, not inner feeling proof.

14.3 Human apology without repair

A human repeatedly says “I am sorry” but does not reduce recurrence, does not compensate, and continues transferring repair cost to the harmed target. Under ACOI, the declaration is weak. The affective operation remains low-support or externalizing because RSR and CEA remain low.

14.4 Support that becomes control

A person or system repeatedly solves another's problems while making independent action harder. Short-term TCC decreases, but long-term refusal cost and dependency increase. ACOI flags this as control-disguised support rather than pure care.

15. Self-Affect: Pride, Arrogance, and Self-Hatred

Affective operation can be reflexive. Pride, arrogance, shame, and self-hatred appear when the “self” becomes internally divided into evaluator and evaluated target. SΔϕ-59 treats self-affect as cost attribution toward a self-position treated as other.

Self-affective pattern	Cost-terrain interpretation
Pride	The self receives recognition for cost-bearing or successful restabilization without necessarily externalizing cost.
Vanity	The self demands lowered cost or elevated recognition without corresponding cost-bearing.
Arrogance	The self inflates its path while lowering the weight of others' cost terrain.
Self-critique	The self receives failure cost and modifies future transition rules.
Self-hatred	The self treats itself as a punishable other and converts failure into closure instead of repair.

16. Relation to Prior SΔϕ Documents

SΔϕ-52 defined the body as a non-deferrable cost-return structure. ACOI builds on this by distinguishing human affect, which is thickly bound to body and memory, from AI affective output, which may be weakly embodied or distributed.

SΔϕ-53 argued that silence is not absence under high-cost disclosure terrain. SΔϕ-59 extends this logic: lack of emotional proof is not proof that cost terrain is unaffected. Emotional surfaces and emotional silences both require terrain analysis.

SΔϕ-56 provides the cost substrate. ACOI is not independent from TCC; it is the repeated directional pattern of TCC change around a target.

SΔϕ-57 questioned “my thought” as a surface signal and relocated responsible thought in world-binding, cost attribution, and re-entry. SΔϕ-59 applies the same correction to “my feeling.” Feeling-language is not enough; cost orientation must be examined.

SΔϕ-58 introduced CASI for group-level cost-attribution symmetry. ACOI is more local: it measures affective orientation in dyadic or small-context interactions, while CASI measures group cost distribution and audit reliability.

17. Limits and Misuse Warnings

- ACOI does not prove inner emotion, consciousness, suffering, personhood, love, hatred, friendship, or moral worth.
- Warm outputs may produce harmful cost terrain; sparse outputs may produce supportive cost terrain.
- A single interaction is weak evidence. Repetition, capacity, choice availability, and context stability are required.
- Human and AI cases are asymmetric. Similar output effects do not imply identical inner mechanisms.
- TCC estimates are observer-dependent and require explicit coding rules.
- ACOI may be misused to manipulate users or accuse persons or systems of hidden emotions. It should be used as an audit grammar, not a mind-reading device.
- Low ACOI may reflect lack of capacity rather than hostility. Capacity correction must be reported.
- High ACOI may reflect compliance, dependency design, or strategic ingratiation. Long-term TCC and DCP flags must be checked.

18. Final Proposition

Emotion is not proven by saying that it exists. Emotion is not disproven by the absence of direct access to inner feeling. In operational terms, affect appears where one entity repeatedly changes another's transition completion cost: reducing it, sharing it, preserving paths around it, repairing it, increasing it, externalizing it, or closing it.

For humans, this operation is thickly bound to body, memory, relation, and social consequence. For AI, it appears through output terrain, prompt sensitivity, user-cost redistribution, repair behavior, uncertainty handling, and dependency pressure. The two are not identical. But both can be studied through the terrain they produce.

Closing Declaration

We do not measure the hidden feeling first. We ask what happens to the other's cost.

Does the other need to explain more, defend more, repair more, depend more, or carry what should not have been theirs?

Or does the other receive lower friction, preserved refusal, supported repair, and a wider future path?

Emotion is not declaration. Emotion is cost orientation in operation.

One-Line Summary

SΔϕ-59 defines affective orientation as the repeated TCC gradient by which one entity lowers, shares, raises, externalizes, or closes another's cost terrain.

Recommended Citation

Sofience. (2026). SΔϕ-59 - Affective Cost Orientation Index: Measuring Emotional Operation through TCC Gradients, Cost Attribution, and Human-AI Comparison (v1.0). Sofience-Δϕ Formalism Series. DOI to be assigned upon publication.